

Fisher Information for First Exposure: Intuition, Geometry, and Practical Uses

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Abstract—Fisher information answers a concrete question: if a model is controlled by an unknown parameter θ , how strongly does the distribution of our observations change when θ changes a little? This note is written for a mathematically mature reader seeing the subject for the first time. We build the concept from the score function, derive the scalar and matrix forms of Fisher information, explain why it yields the Cramér-Rao bound and a natural Riemannian metric, and then emphasize how practitioners actually use it. The main practical uses are: predicting estimation accuracy before data are collected, diagnosing non-identifiability, choosing experiments and sensor configurations, and preconditioning optimization in machine learning. The final sections collect literature-backed examples from microscopy, pharmacodynamics, MRI, robotics, and AI, and then separate those established uses from forward-looking engineering ideas.

Index Terms—Fisher information, Cramér-Rao bound, information geometry, experimental design, robotics, machine learning, natural gradient

version: 2026-03-28-14:05:55-CET-a7150d3-main

I. INTRODUCTION

Fisher information is one of those ideas that first looks technical and then turns out to be surprisingly concrete. It is about *distinguishability*. If two nearby parameter values produce almost the same distribution of observations, then the data do not tell us much about which parameter value is true. If two nearby parameter values produce noticeably different distributions, then the data are informative.

This simple idea has several consequences:

- 1) It gives a local measure of how informative an observation is about a parameter.
- 2) It adds across independent samples, so it connects directly to sample size.
- 3) Its inverse sets the natural lower bound on estimator covariance.
- 4) It acts as a metric tensor on parameter space, which is why geometry enters.
- 5) It becomes a design tool: one can choose experiments, sensors, or trajectories to maximize information.

This document is intentionally written as a first encounter rather than as a compressed review. I will therefore move slowly, beginning with the scalar case, then the matrix case, then the geometric interpretation, and only afterwards the more modern robotics and AI uses.

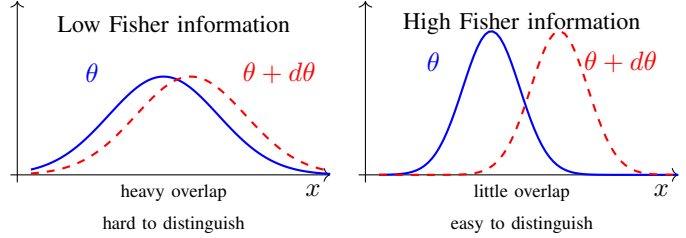


Fig. 1. Same idea in pictures: Fisher information is large when a small parameter change produces a noticeable change in the observation distribution.

If you remember only one sentence, remember this:
Fisher information is *local likelihood sharpness measured in parameter units*.

II. A FIRST PICTURE

Suppose $X \sim p(x | \theta)$, where θ is unknown. We want to know how much one observation of X can tell us about θ .

Take two nearby parameter values, θ and $\theta + d\theta$. If the two distributions $p(x | \theta)$ and $p(x | \theta + d\theta)$ heavily overlap, then an observed value x does not help much in telling them apart. If the distributions are well separated, then even one sample is informative.

Figure 1 is already most of the story. The rest of the theory formalizes this local distinguishability in a way that supports estimation, uncertainty quantification, geometry, and design.

III. SCALAR FISHER INFORMATION

A. The Score Function

To measure how sensitive the model is to θ , we differentiate the log-likelihood:

$$S_\theta(X) = \frac{\partial}{\partial \theta} \log p(X | \theta).$$

This random variable is called the *score*. For one realized observation x , the score says how strongly the log-likelihood wants to push θ upward or downward at the current parameter value.

Under standard regularity conditions (smoothness of $p(x | \theta)$, common support, and permission to interchange differentiation and integration),

$$\mathbb{E}_\theta[S_\theta(X)] = 0.$$

The proof is short:

$$\mathbb{E}_\theta[S_\theta(X)] = \int \frac{\partial_\theta p(x | \theta)}{p(x | \theta)} p(x | \theta) d\mu(x) \quad (1)$$

$$= \int \partial_\theta p(x | \theta) d\mu(x) \quad (2)$$

$$= \partial_\theta \int p(x | \theta) d\mu(x) = \partial_\theta 1 = 0. \quad (3)$$

So the score has mean zero. That matters because it means its variance is the same thing as its second moment.

B. Definition

Definition. The scalar Fisher information is

$$I(\theta) = \mathbb{E}_\theta[S_\theta(X)^2] = \text{Var}_\theta(S_\theta(X)).$$

There is also an equivalent curvature form:

$$I(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \log p(X | \theta) \right].$$

These two formulas are worth interpreting separately:

- 1) $I(\theta) = \text{Var}(S_\theta)$: Fisher information is the typical *size* of the score. Large score fluctuations mean the data react strongly to parameter changes.
- 2) $I(\theta) = -\mathbb{E}(\ell''(\theta))$: Fisher information is the expected *curvature* of the log-likelihood. A sharp likelihood peak means high information.

Working interpretation. The score tells you the local direction in which the data want the parameter to move. Fisher information tells you how strongly the data care.

C. Why the Square Appears

Beginners often wonder why Fisher information uses the *square* of the score. The answer is simple: the sign of the score depends on the observed sample and averages to zero. Squaring removes the sign and measures the typical magnitude of the local sensitivity.

If the score is almost always tiny, the model barely changes when θ changes, so information is low. If the score is typically large in magnitude, then the model is sensitive to θ , so information is high.

D. Independent Samples Add

If $X_1, \dots, X_n$ are i.i.d. from $p(x | \theta)$, then the log-likelihood adds:

$$\ell_n(\theta) = \sum_{i=1}^n \log p(X_i | \theta).$$

Therefore the score adds:

$$S_{\theta,n} = \sum_{i=1}^n S_\theta(X_i),$$

and, because the scores are independent and mean zero,

$$I_n(\theta) = n I_1(\theta).$$

This is one of the most useful facts in practice. Fisher information scales linearly with sample size, and uncertainty therefore scales like $1/\sqrt{n}$.

IV. THREE WORKED ONE-PARAMETER EXAMPLES

A. Gaussian Mean with Known Variance

Let $X \sim \mathcal{N}(\mu, \sigma^2)$, where σ^2 is known and μ is unknown. Then

$$\log p(x | \mu) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x - \mu)^2}{2\sigma^2}.$$

Differentiate with respect to μ :

$$S_\mu(x) = \frac{x - \mu}{\sigma^2}.$$

Hence

$$I(\mu) = \mathbb{E} \left[\frac{(X - \mu)^2}{\sigma^4} \right] = \frac{\sigma^2}{\sigma^4} = \frac{1}{\sigma^2}.$$

This is an ideal first example because the interpretation is clean:

- smaller measurement noise σ^2 means larger information,
- n samples give n/σ^2 information,
- any unbiased estimator of μ must have variance at least σ^2/n .

B. Bernoulli Probability

Let $X \sim \text{Bernoulli}(p)$, so

$$p(X = x | p) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

Then

$$\log p(x | p) = x \log p + (1 - x) \log(1 - p),$$

so

$$S_p(x) = \frac{x}{p} - \frac{1 - x}{1 - p} = \frac{x - p}{p(1 - p)}.$$

Because $\text{Var}(X) = p(1 - p)$,

$$I(p) = \mathbb{E} \left[\frac{(X - p)^2}{p^2(1 - p)^2} \right] = \frac{1}{p(1 - p)}.$$

This surprises many people the first time they see it: $I(p)$ is *largest* near $p = 0$ or $p = 1$, not at $p = 1/2$. On the absolute p -scale, a near-deterministic coin is easier to estimate than a fair coin because the observation variance is smaller near the boundary.

This is also a good reminder that parameterization matters numerically. If we reparameterize by the log-odds

$$\eta = \log \frac{p}{1 - p},$$

then the Fisher information with respect to η becomes $I(\eta) = p(1 - p)$, which is maximal at $p = 1/2$.

TABLE I
THREE FIRST EXAMPLES. HERE I_1 MEANS FISHER INFORMATION FROM ONE OBSERVATION, AND THE LAST COLUMN GIVES THE SCALAR CRAMÉR-RAO LOWER BOUND FOR n I.I.D. OBSERVATIONS.

Model	Parameter	I_1	$\text{Var}(\hat{\theta}) \geq 1/(nI_1)$
$\mathcal{N}(\mu, \sigma^2)$	μ	$1/\sigma^2$	σ^2/n
Bernoulli(p)	p	$1/[p(1-p)]$	$p(1-p)/n$
Poisson(λ)	λ	$1/\lambda$	λ/n

C. Poisson Rate

Let $X \sim \text{Poisson}(\lambda)$, $\lambda > 0$. Then

$$\log p(x | \lambda) = x \log \lambda - \lambda - \log(x!),$$

so

$$S_\lambda(x) = \frac{x}{\lambda} - 1.$$

Since $\text{Var}(X) = \lambda$,

$$I(\lambda) = \text{Var}\left(\frac{X}{\lambda}\right) = \frac{1}{\lambda}.$$

So one Poisson count contains less information about a large rate λ than about a small rate, when the error is measured on the absolute λ -scale.

D. A Useful Summary Table

V. FROM A NUMBER TO A MATRIX

A. Vector Parameters

Now let $\theta = (\theta_1, \dots, \theta_d) \in \mathbb{R}^d$. The scalar score becomes the *score vector*

$$s_\theta(X) = \nabla_\theta \log p(X | \theta),$$

and the scalar Fisher information becomes the *Fisher information matrix*

$$I(\theta) = \mathbb{E}_\theta [s_\theta(X) s_\theta(X)^\top].$$

Componentwise,

$$I_{ij}(\theta) = \mathbb{E}_\theta \left[\frac{\partial}{\partial \theta_i} \log p(X | \theta) \frac{\partial}{\partial \theta_j} \log p(X | \theta) \right].$$

The matrix interpretation is straightforward:

- diagonal entries measure information about each parameter separately,
- off-diagonal entries measure coupling between parameters,
- small eigenvalues indicate weakly identifiable directions,
- large eigenvalues indicate well-observed directions.

B. Expected Information and Observed Information

In practice one also sees the *observed information*,

$$J_{\text{obs}}(\hat{\theta}; x) = -\nabla_{\hat{\theta}}^2 \ell(\theta; x)|_{\theta=\hat{\theta}}.$$

This is the negative Hessian of the realized log-likelihood at an estimate $\hat{\theta}$. It depends on the data sample x , while the Fisher information matrix is the *expectation* of that local curvature under the model.

This distinction matters in applications:

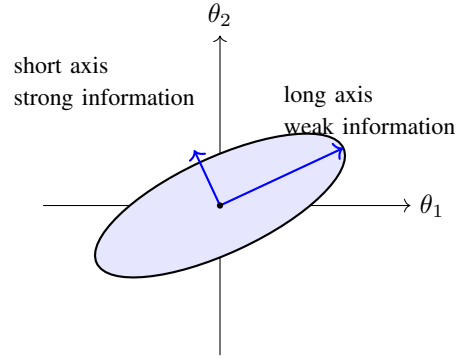


Fig. 2. In multiple dimensions, the inverse Fisher matrix defines a local uncertainty ellipse. Long directions are poorly informed directions.

- theoretical derivations usually use expected Fisher information,
- optimization software often returns observed information,
- deep learning papers often use approximate or empirical versions of the Fisher matrix, which are related but not identical.

C. A Two-Parameter Example

For $X \sim \mathcal{N}(\mu, \sigma^2)$ with parameter vector (μ, σ^2) ,

$$I(\mu, \sigma^2) = \begin{bmatrix} \sigma^{-2} & 0 \\ 0 & (2\sigma^4)^{-1} \end{bmatrix}.$$

The zero off-diagonal entry tells us that, in this parameterization, mean and variance decouple locally. The larger diagonal entry corresponds to the direction that is easier to estimate.

Figure 2 is a useful mental picture. Fisher information is not just “how much” information we have. It also tells us *in which directions* of parameter space the information lies.

VI. WHY FISHER INFORMATION CONTROLS ESTIMATION ERROR

A. The Cramér-Rao Bound

If $\hat{\theta}(X)$ is an unbiased estimator of a scalar parameter, then

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}.$$

For n i.i.d. samples,

$$\text{Var}(\hat{\theta}) \geq \frac{1}{nI_1(\theta)}.$$

In the vector case, the matrix form is

$$\text{Cov}(\hat{\theta}) \succeq I(\theta)^{-1},$$

where $\succeq$ denotes positive semidefinite ordering.

This bound matters for two reasons:

- 1) it tells us the best covariance we could hope for, before we design an estimator,
- 2) it turns $I(\theta)^{-1}$ into a natural local scale for parameter uncertainty.

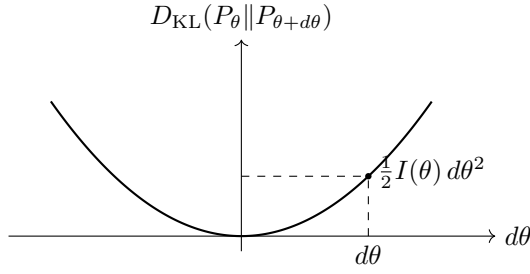


Fig. 3. Locally, KL divergence looks quadratic in the parameter displacement. The quadratic coefficient is the Fisher information.

B. What the Bound Really Says

The Cramér-Rao bound does *not* say that every estimator reaches this lower bound. It says that, under the relevant assumptions, no unbiased estimator can do better. In many regular models, the maximum likelihood estimator is asymptotically efficient, meaning that for large samples it approaches this lower bound.

That is why practitioners often use Fisher information before collecting data: it predicts the *best-case local accuracy* of any reasonable estimator.

VII. WHY GEOMETRY APPEARS

A. Local KL Divergence

The geometric interpretation comes from a second-order expansion of the Kullback-Leibler divergence. For nearby parameters,

$$D_{\text{KL}}(P_\theta \parallel P_{\theta+d\theta}) = \frac{1}{2} d\theta^\top I(\theta) d\theta + O(\|d\theta\|^3).$$

This is the key bridge from statistics to geometry. It says that, to second order, Fisher information is the quadratic form governing local distinguishability.

B. The Fisher-Rao Metric

Because $d\theta^\top I(\theta) d\theta$ is a positive quadratic form, it behaves like a Riemannian line element:

$$ds^2 = d\theta^\top I(\theta) d\theta.$$

This is the *Fisher-Rao metric*.

The geometry is not decoration. It says that parameter space is not naturally Euclidean from the viewpoint of inference. A unit change in one coordinate may matter far more than a unit change in another, and that anisotropy is exactly what the Fisher matrix captures.

It is worth being careful here. Geodesics of the Fisher-Rao metric are the shortest paths in this information-based geometry. They are *not* automatically the trajectory followed by an estimator or optimizer. The geometry tells us how the statistical model is shaped, not what a specific algorithm must do.

C. A Useful Geometric Algebra Parallel

If you instinctively think in geometric algebra, there is a clean and honest connection here.

At each point θ , the Fisher matrix defines an inner product on the tangent space:

$$g_\theta(u, v) = u^\top I(\theta) v.$$

Once an inner product is available, one can build the corresponding Clifford algebra

$$\text{Cl}(T_\theta \Theta, g_\theta).$$

So a statistical manifold equipped with the Fisher-Rao metric automatically carries a *pointwise family of geometric algebras*, just as any Riemannian manifold does.

That observation gives several good parallels:

- the score $d \log p$ is naturally a covector field,
- the natural gradient is what you get after using the metric $I(\theta)^{-1}$ to convert that covector into a vector,
- uncertainty ellipsoids are the metric balls induced by g_θ ,
- oriented parameter-area elements can be thought of as bivectors on the tangent space.

The important caveat is that this is mostly a *structural connection*, not standard notation in the Fisher-information literature. In classical statistics, people usually stop at tensors, quadratic forms, and Riemannian geometry; they rarely continue into full geometric-algebra language. There are niche examples in quantum information geometry where geometric algebra and information geometry are used together, but that is not the mainstream classical viewpoint.

VIII. HOW YOU ACTUALLY USE FISHER INFORMATION

A. A Practical Recipe

Suppose your experiment or system has a design variable u : sample time, sensor placement, control trajectory, excitation signal, camera pose, coil arrangement, and so on. Then Fisher information becomes $I(\theta; u)$, and a very common workflow is:

- 1) write a measurement model $p(y \mid \theta, u)$,
- 2) compute or approximate the Fisher matrix,
- 3) decide what “good” means,
- 4) optimize u subject to practical constraints.

Three standard scalar summaries are:

$$\text{D-optimality:} \quad \max_u \det I(\theta; u), \quad (4)$$

$$\text{A-optimality:} \quad \min_u \text{tr} (I(\theta; u)^{-1}), \quad (5)$$

$$\text{E-optimality:} \quad \max_u \lambda_{\min} (I(\theta; u)). \quad (6)$$

Each criterion emphasizes a different notion of “best”:

- D-optimality minimizes the volume of the local uncertainty ellipsoid,
- A-optimality minimizes average variance,
- E-optimality protects the weakest identifiable direction.

B. A/B Tests and Sample Size

In a Bernoulli model, Fisher information gives the best-case variance scale $p(1-p)/n$ for estimating a conversion rate. This is the same $1/\sqrt{n}$ logic behind sample-size calculations: double the number of independent samples and you improve standard errors only by about a factor of $\sqrt{2}$.

C. Sensor Fusion

If two independent sensors both measure the same scalar parameter, their information adds. For example, if

$$X_A \sim \mathcal{N}(\theta, \sigma_A^2), \quad X_B \sim \mathcal{N}(\theta, \sigma_B^2),$$

then

$$I_{\text{total}}(\theta) = \frac{1}{\sigma_A^2} + \frac{1}{\sigma_B^2}.$$

This is exactly the right intuition: precise sensors contribute more, noisy sensors contribute less, and combining independent evidence adds information.

D. Diagnosing Non-Identifiability

If the Fisher matrix is singular or nearly singular, something is wrong from an inference perspective:

- two parameters may be redundant,
- the experiment may not excite the relevant dynamics,
- the sensor may not observe some directions of the state or parameter space.

Practically, inspecting the eigenvalues and eigenvectors of $I(\theta)$ is a powerful debugging tool. It tells you not just that the problem is ill-conditioned, but *which combinations of parameters* are poorly determined.

E. Natural Gradient and Optimization

In Euclidean optimization one takes a step proportional to $-\nabla L(\theta)$. But if parameter space has a natural statistical geometry, the Euclidean gradient is coordinate-dependent in a bad way. The *natural gradient* corrects this by using

$$\Delta\theta \propto -I(\theta)^{-1} \nabla L(\theta).$$

This rescales the update by the local geometry of the model family. In machine learning, this idea is especially useful when different parameters have very different statistical sensitivities [1], [2], [3], [4].

IX. EXAMPLES FROM PRACTICE

This section answers the direct question “where is Fisher information actually used?” The common pattern is easy to spot: Fisher information is most useful when someone must estimate parameters *and* can influence how the data will be collected.

A. Single-Molecule Microscopy

Ober, Ram, and Ward used the Fisher information matrix to derive the localization accuracy limit for a single fluorescent molecule in microscopy [5]. The use is very concrete: the Fisher matrix predicts the best achievable localization precision, and the paper studies how pixelation and detector noise degrade that precision. This is a classic example of Fisher information as a performance limit *and* as a design guide.

B. Population Pharmacodynamics

Ogunbenro and Aarons used the population Fisher information matrix to optimize experimental design for population pharmacodynamic studies with discrete responses [6]. Here Fisher information tells the experimenter which sampling schedules and response structures are most informative for identifying model parameters. This is a clean example of Fisher information as an engineering tool for deciding *when to measure*.

C. Quantitative MRI and Parallel Imaging

Bouhrara and Spencer used Fisher information and the Cramér-Rao bound for experimental design in parallel MRI, with careful attention to the correct noise model [7]. The point is not abstract: different acquisition choices imply different Fisher matrices, and therefore different achievable uncertainty in estimated tissue parameters.

D. Sensor Configuration and Sensor Management

Williams, Suvorov, Wang, and Moran explicitly study sensor configuration through the Fisher matrix and information geometry [8]. Their setup treats sensor placement or reconfiguration as an optimization problem over information content. This is a very general design pattern, and it appears again and again in tracking, localization, and active sensing.

X. ROBOTICS AND AI

This section is split in two parts. The first two subsections are literature-backed uses. The last subsection is intentionally speculative: those ideas are my inferences from the cited work, not claims that a specific paper has already demonstrated them.

A. Established Uses in Robotics

Extrinsic sensor calibration. Brookshire and Teller formulate sensor-to-sensor calibration from egomotion measurements using a constrained Fisher information matrix and a corresponding Cramér-Rao lower bound [9]. The Fisher matrix is used both to quantify achievable calibration accuracy and to analyze *motion degeneracy*: some robot motions simply do not excite enough information to identify the calibration robustly.

Information-maximizing excitation trajectories. Wilson, Schultz, and Murphey optimize trajectories to improve parameter identification in nonlinear dynamical systems by maximizing a norm of the Fisher information matrix [10]. Their experiments show a large increase in the smallest eigenvalue of the Fisher matrix and a corresponding reduction in estimation error. This is Fisher information used as a direct *control objective*.

Perception-aware planning and localization. Zhang and Scaramuzza build a Fisher Information Field for visual localization and motion planning [11]. Their idea is to precompute location-dependent Fisher information in a map so that planners can quickly prefer trajectories that are easier to localize. This is a particularly nice robotics use case because it turns an estimation quantity into a reusable planning primitive.

B. Established Uses in AI and Machine Learning

Natural gradient learning. Amari’s natural gradient paper is the canonical starting point [1]. The core idea is to precondition gradient steps by the inverse Fisher matrix so that optimization respects the geometry of the model family rather than the arbitrary coordinates chosen for the parameters.

Reinforcement learning. Kakade’s natural policy gradient applies this geometric idea to policy optimization [3]. The Fisher matrix of the policy distribution determines the local notion of steepest ascent in policy space.

Scalable second-order optimization for deep nets. Martens and Grosse’s K-FAC method approximates the Fisher matrix in a structured way so that one can get much of the benefit of natural-gradient-style updates without forming the full matrix explicitly [2]. This is Fisher information as an optimization preconditioner at deep-learning scale.

Black-box optimization and evolution strategies. Wierstra and collaborators use the natural gradient of a search distribution in Natural Evolution Strategies [4]. The parameters being updated are the parameters of the search distribution itself, so the Fisher matrix lives over the family of candidate-search distributions.

Continual learning. Kirkpatrick and collaborators use a diagonal approximation to the Fisher matrix in Elastic Weight Consolidation (EWC) [12]. The intuition is that large Fisher entries indicate parameters that strongly matter to previously learned tasks, so changing them should be penalized when learning a new task.

C. Promising Directions for Robotics and AI (Inference, Not Literature Claims)

The papers above suggest several directions that seem genuinely promising:

Task-aware active SLAM. Instead of maximizing information about the robot pose alone, one could optimize a Fisher objective over the task-relevant latent variables: object poses to manipulate, terrain properties relevant to locomotion, or contact parameters relevant to insertion and grasping.

Sensor scheduling for embodied AI. In multimodal robots, Fisher information could help decide when to turn on expensive sensors. If the Fisher gain from activating a depth sensor, tactile array, or high-rate camera is small in the current regime, the robot could save power and bandwidth.

Data collection for robot learning. In system identification or sim-to-real adaptation, trajectories could be chosen to maximize information about the parameters that dominate downstream policy performance, rather than about all model parameters equally.

Uncertainty-aware world-model debugging. Small eigenvalues of a Fisher matrix or of a closely related sensitivity matrix can expose directions in parameter space that the current training data do not constrain. For large robot world models, this could become a concrete debugging signal for missing coverage.

Fisher-guided continual adaptation in embodied agents. EWC suggests an obvious robotics extension: use Fisher-

derived parameter importance to let a deployed robot adapt online without overwriting the parts of its controller or perception stack that are critical for safety.

XI. COMMON PITFALLS AND FIRST-EXPOSURE CONFUSIONS

A. Fisher Information Is Local

Fisher information only describes *local* distinguishability around the current parameter value. Two distant parameter values can be globally easy or hard to tell apart in ways that a local quadratic approximation does not fully capture.

B. It Depends on the Model

The Fisher matrix is only as good as the statistical model $p(x | \theta)$. If the model is badly misspecified, then the Fisher matrix may still quantify curvature of the chosen likelihood, but that curvature may not describe the real world well.

C. Parameterization Changes the Numbers

The raw matrix entries change under reparameterization. That is not a bug. It is exactly how a metric tensor behaves. The *geometry* is invariant, but the coordinate expression is not.

D. Expected, Observed, and Empirical Fisher Are Different

These are often conflated:

- **expected Fisher:** model expectation of score outer products,
- **observed Fisher:** realized negative Hessian at the data,
- **empirical Fisher:** data-average of per-sample score outer products.

They are related, but they are not interchangeable in every argument.

E. Boundary and Support Issues Matter

Some identities used in clean textbook derivations fail when the support of the distribution depends on the parameter, or when regularity assumptions break. This is not a minor technicality. It changes what formulas are valid.

XII. CONCLUSION

Fisher information is a single object with several faces:

- a variance of the score,
- a curvature of the log-likelihood,
- a lower-bound generator through the Cramér-Rao inequality,
- a metric tensor for statistical geometry,
- and a design criterion for experiments, sensors, trajectories, and optimizers.

For a first mental model, the right one is still the simplest: Fisher information measures how much a small change in the parameter changes what you expect to observe. Once that idea is internalized, the rest of the subject becomes much less mysterious.

If I had to summarize the practical lesson in one line, it would be this: *compute Fisher information whenever you want to know, before running the whole pipeline, whether your data-collection or optimization setup is even capable of identifying what you care about.*

SCOPE NOTE

All concrete application claims in Sections VIII and IX are tied to cited papers. The final subsection of Section IX is deliberately marked as inference: those are proposed directions suggested by the cited literature, not claims of already-established results.

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